

SymPy Bug Card

Bug 35: solveset includes negative real non-solutions

Task. Solve over the complex numbers:

$$\sqrt{x}\sqrt{1/x} = 1.$$

SymPy's answer.

$$\text{solveset}\left(\sqrt{x}\sqrt{1/x} = 1, x, \mathbb{C}\right) = \mathbb{C} \setminus \{0\}.$$

Correct answer.

$$\mathbb{C} \setminus \{x \in \mathbb{R} : x \leq 0\}.$$

Explanation. The returned set contains $x = -1$, but

$$\sqrt{-1}\sqrt{1/(-1)} = i \cdot i = -1 \neq 1.$$

The principal complex square root is not multiplicative across the negative real branch cut, so the negative real axis must be excluded.

Likely root cause. In `sympy/solvers/solveset.py`, `_solve_radical` accepts an infinite superset produced by `unrad` without rechecking it against the original radical equation. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-035-solveset-includes-negative-real-non-solutions/](#)